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Course:computer networks
Course code:CSA0704
Model lab

SET 7

1. Design a multimedia network capable of streaming audio and video to multiple devices simultaneously using Packet Tracer.
2. Set up AAA (Authentication, Authorization, and Accounting) using both local and server-based methods in an IoT environment using Packet Tracer.
3. Using Wireshark, Capture and analyze the Packets that uses IP for different header fields and payload of the concerned packet.
4. Set up a basic network using three PCs and two switches, ensuring all devices can communicate with each other.

Model Practical

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1. Aim:

To design a multimedia network using Cisco packet Tracer that can stream audio and video content from a Server to multi client devices.

Requirement:

- 1 router (e.g.: 1941)
- 1 switch (e.g.: 2960)
- 1 Server
- 3 PCs
- Copper straight-through Cables

Procedure:

Step 1: Open Cisco packet Tracer

Step 2: Place the devices.

Step 3: Connect the devices

Step 4: Configure the Server router

Step 5: Configure the Server.

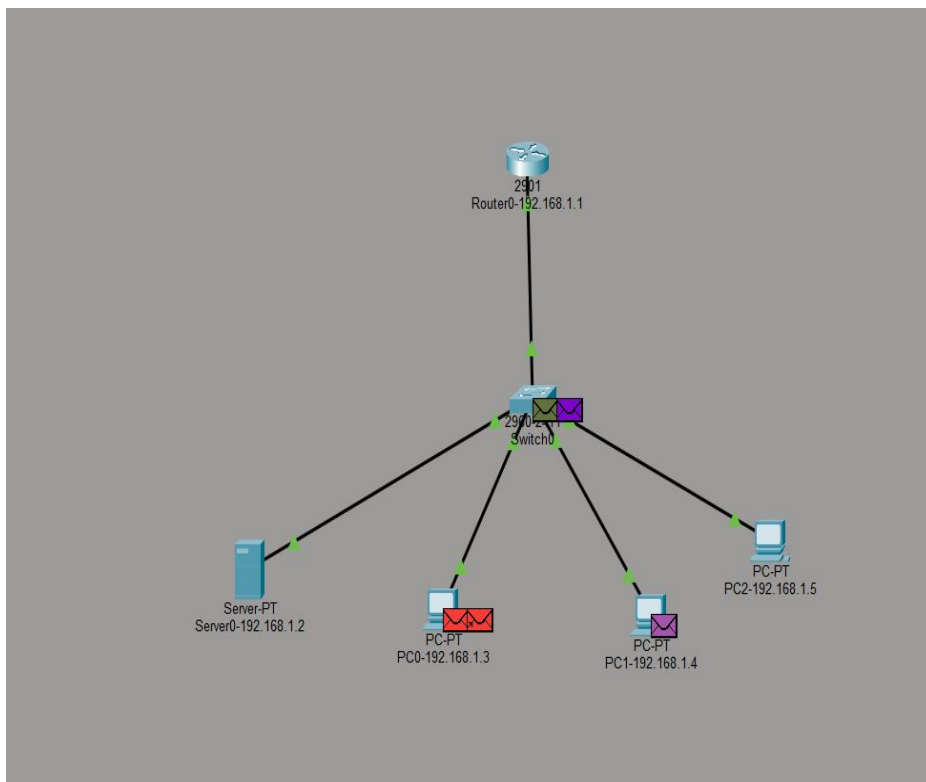
Step 6: Configure PC's

Step 7 → Test Connectivity

Step 8 → Test multimedia Server access.

Result:

A multimedia network was successfully designed in packet tracer. All PC's can access Server and stream audio/video content.



2. Aim:
To Configure and test AAA using both local authentication and server-based authentication on a router.

Device Required:

- 1 Router (R1)
- 1 Switch
- 1 AAA Server
- 1PC [For testing Telnet/SSH]

Procedure:

- Step 1: open Cisco packet tracer.
- Step 2: Place the devices
- Step 3: Connect the devices
- Step 4: configure IP Address
- Step 5: Configure Routers
- Step 6: Configure Local AAA and Server-based AAA
- Step 7: Configure RADIUS on Router
- Step 8: Apply Server AAA to VTY
- Step 9: Test Server-based AAA

Result:

AAA was successfully configured using both local and server-based methods and verify using telnet.

Router1
(R1) Router1
Admin E
Router AAA E

[H22 \telnet prompt test] OK

: successful

Router1+telnet OK OK OK OK

Router1 OK OK OK OK

Router1 OK OK OK OK

Router1 OK OK OK OK

Router1 OK OK OK OK

Router1 OK OK OK OK

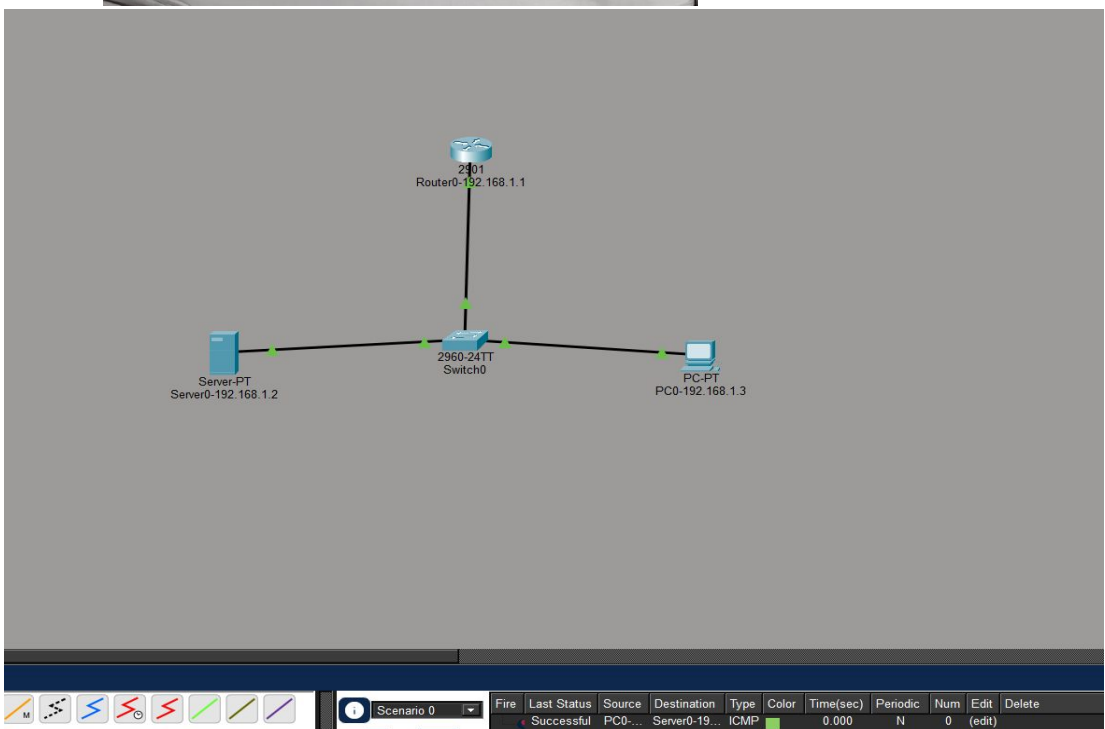
AAA local OK OK OK

Router1 OK OK OK OK

Router1 OK OK OK OK

Router1 OK OK OK OK

Router1 OK OK OK OK



Aim:
To capture IPV4 packet using
Wireshark and analyze various IP
header field and payload contents

Material Required:

- Wireshark Network Analyzer
- PC connected to active network
- Command prompt/Terminal

Procedure:

Step-1 — Start Packet Capture

Step-2 — Generate Traffic

Step-3 — Apply Display Filter

Step-4 — Analyze IP Header fields

Step-5 — Analyze Payload

Result:

IPV4 packet were successfully
captured using Wireshark. and their
IPV4 header fields along with inner
payload structure were thoroughly
analyzed.

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The main window is divided into three panes: the packet list, packet details, and packet bytes. The packet list shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, and Length. The selected packet is an Internet Protocol Version 4 (IP) packet, and its details pane shows the IP header fields and the payload structure. The packet bytes pane shows the raw data of the selected packet.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
6480	168.380432100	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=51209 Win=4193024 Len=0
6481	168.380432500	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=52649 Win=4194560 Len=0
6482	168.380432800	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=54089 Win=4193024 Len=0
6483	168.380433300	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=55529 Win=4194560 Len=0
6484	168.383830900	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=58489 Win=4194560 Len=0
6485	168.383831700	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=59849 Win=4193024 Len=0
6486	168.383832000	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=61289 Win=4194560 Len=0
6487	168.383832500	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=67049 Win=4194560 Len=0
6488	168.383832900	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=71369 Win=4194560 Len=0
6489	168.383833100	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=9194 Ack=74495 Win=4194560 Len=0
6495	168.958937600	172.25.104.195	224.77.77.77	UDP	155	53808 → 12177 Len=113
6496	168.958938500	172.25.106.175	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
6498	168.959574000	172.25.106.175	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
6502	169.250577000	13.107.139.11	172.25.108.49	TCP	1514	443 → 55257 [ACK] Seq=9194 Ack=74495 Win=4194560 Len=1460
6503	169.250587100	13.107.139.11	172.25.108.49	TCP	1514	443 → 55257 [ACK] Seq=10654 Ack=74495 Win=4194560 Len=1460 [TCP PDU reassembled in 6504]
6504	169.250588500	13.107.139.11	172.25.108.49	TLSv1.3	114	Application Data
6505	169.250591800	13.107.139.11	172.25.108.49	TCP	1514	[TCP Retransmission] 443 → 55257 [PSH, ACK] Seq=10714 Ack=74495 Win=4194560 Len=1460 [TCP PDU reassembled in 6504]
6506	169.250748000	172.25.108.49	13.107.139.11	TCP	66	55257 → 443 [ACK] Seq=74495 Ack=12174 Win=65280 Len=0 SLE=10714 SRE=12174
6507	169.327648900	172.25.108.49	13.107.139.11	TLSv1.3	2850	Application Data
6508	169.351015100	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=12174 Ack=75935 Win=4193024 Len=0
6509	169.351015600	13.107.139.11	172.25.108.49	TCP	54	443 → 55257 [ACK] Seq=12174 Ack=77291 Win=4194560 Len=0
6510	169.877738200	172.25.105.80	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _spotify-connect._tcp.local, "QM" question

Packet Details:

Frame 2663: Packet, 153 bytes on wire (1224 bits), 153 bytes captured (1224 bits) on interface \Device\NPF...
Ethernet II, Src: CloudNetwork 86:03:53 (08:f9:7e:86:03:53), Dst: Cisco6d:c0:e7 (24:d5:e4:6d:c0:e7)
Internet Protocol Version 4, Src: 172.25.108.49, Dst: 4.213.25.242
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 139
Identification: 0x6d21 (27937)
0000 = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 172.25.108.49
Destination Address: 4.213.25.242
[Stream index: 103]
Transmission Control Protocol, Src Port: 49535, Dst Port: 443, Seq: 99, Ack: 170, Len: 99
Transport Layer Security

Packet Bytes:

0000 24 d5 e4 6d c0 e7 00 f9 7a 86 03 53 00 00 85 00
0010 00 8b 6d 21 00 00 00 06 00 00 ac 19 66 31 04 d5
0020 19 f2 c1 7f 01 bb 0c 14 19 53 b8 27 6d a3 50 18
0030 00 fe f3 e3 00 00 17 03 03 00 5e 00 00 00 00 00
0040 00 00 09 df 1d c7 cb fe 83 c6 e3 04 7c 67 e1 0
0050 ba 1e a0 74 c6 db 40 1f 65 ba 8f dc 0c 13 b4 4
0060 5d 19 79 00 52 28 3d 34 cf 68 04 19 95 23 40 9
0070 9b ea 97 30 92 24 f1 de fd 46 02 13 75 2d 27 c
0080 fd dc b1 20 7a 58 a9 e1 9b 8c c5 56 23 a4 a7 9
0090 bd fa a3 24 23 ce 0d 5d ba

Sniffing Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Mark-up and share

4

Aim:

To design and configure a basic network topology connecting three PCs and two switches to ensure full inter-device communication using Cisco Packet Tracer.

Material Required:

- 3 PC's
- Switches
- Copper Straight Cables

Procedure:

Step 1 - Build Topology

Step 2 - Configure IP Address

Step 3 - Verify Connectivity

Result:

Successful ICMP reply message were received from all three target IP address.

